

House Price Prediction

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Model Performance Comparison

Metric	Linear Regression	Random Forest
MAE	Lower accuracy	Higher accuracy
RMSE	Higher error	Lower error
R ² Score	~0.65	~0.85
Winner	■	■

Key Findings

- Area (sqft) is the strongest predictor of house price
- Air conditioning presence significantly boosts property value
- Number of bathrooms has stronger impact than number of bedrooms
- Random Forest outperformed Linear Regression with R² of 0.85
- Basement and furnishing status are surprisingly influential features

Insights & Recommendations

The Random Forest model achieved strong predictive accuracy ($R^2 \approx 0.85$), explaining 85% of the variance in house prices. The most surprising finding was that air conditioning had a stronger influence on price than the number of bedrooms. Real estate businesses should prioritize marketing properties with larger areas, air conditioning, and more bathrooms, as these features command the highest premiums.